

Tianyuan (Tiana) Wang

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EDUCATION

University of Georgia

PhD in Agricultural and Applied Economics (Committee Chair: Dr. Michael Adjemian)
M.S. in Statistics

Athens, GA

Sep 2023 – Present

Sep 2023 – Present

Johns Hopkins University

M.S. in Government Analytics (Data Analytics and Policy)

Washington, DC

Sep 2018 – Jun 2020

Hong Kong Baptist University

B.B.A. in Applied Economics with a minor in Finance

Hong Kong

Sep 2014 – Jun 2018

SKILLS & INTERESTS

Skills: AI Native, R, [R Shiny](#), Python, SQL, Stata, $\text{\LaTeX}$, Microsoft Suite, PowerBI, Tableau

Expertise: Food Prices, Cost Provision, Big Data Integration

Hobbies: Swimming (Certified as China National Level II Athlete), Tennis, Hiking, Foodies

WORKING EXPERIENCE

International Monetary Fund (IMF)

Research Analyst (Outstanding Performance, top 10 under 30% curve distribution)

Washington, DC

Dec 2020 – Jul 2023

- Closely working with economists in three SACU country teams and across four departments, specializing in research support, ETL pipelines, and timely resolution of glitches. [Excel, STATA, R]
- Developed and regularly maintained interactive surveillance dashboards for economic diagnosis, and drafted data-driven analyses with charts for IMF Article IV Staff Reports. [Excel, Python, R, Flexdashboard in R]
- Managed economic outlook files, including cleaning, restructuring, and modeling for four sectors (real, fiscal, external, financial) to enhance efficiency and consistency; collaborated with country teams and national authorities to update data and projections. [Excel, SQL]
- Periodically appraised inputs and outputs of historical series and 5-year outlooks, validated indicator calculations using cross-country datasets, and developed a validation process for internal data submissions. [Excel]
- Drafted contingency memos on topics including civil unrest in South Africa and Eswatini, the COVID-19 situation in Namibia, and the Namibia Policy Responses Tracker to COVID-19 (details in [COVID-19 IMF Policy Tracker](#)).

Global Risk Intelligence

Lead Data Scientist

Washington, DC

Jun 2019 – Feb 2020

- Led a team of 3 colleagues to design and implement a machine learning project focused on supply chain risk in the aerospace industry; presented findings and strategic recommendations to high-profile clients. [R, Tableau]
- Built a data pipeline using decision-tree classification to provide risk-based recommendations for diversifying procurement sources and clustering analysis to evaluate similarities among suppliers. [Python]
- Performed various NLP tasks, including text mining from Twitter and public-accessible documents; conducted keyness testing (Quanteda in R) with sentiment analysis (VADER in Python), and implemented Structured Topic Modeling (STM) to assess supplier similarities. [R, Python, Excel]
- Developed a risk assessment dashboard with a structured playbook to ensure reproducibility. [R, Tableau]

PUBLICATION

Albertin, G., Kazandjian, R., Thioune, F. K., & Wang, T. T. (2024). Chapter 4: COVID-19 and Gender Inequality: Impact in Southern Africa. *Gender Equality and Economic Development in Sub-Saharan Africa*. USA: International Monetary Fund.

WORKING IN PROGRESS

“Food Price Inflation in the United States” with [Dr. Michael Adjemian](#) and [Qingxiao Li](#)

CONFERENCE PRESENTATIONS

SAEA 2026 Annual Meeting (Louisville, KY), AAEA 2026 Annual Meeting (Kansas City, MO)

A Machine Learning-based Toolbox for Climate Policy Analysis

Washington, DC

\$50,000 grant from International Monetary fund | [Brief](#)

Research Analyst & Toolbox Developer

Oct 2021 – Jul 2023

- Led the exploration of potential analysis methods, ensuring full consideration of data availability and model feasibility using the Climate Policy Database; prepared the proposal and successfully pitched the project. [R]
- Developed a machine learning prototype in R to perform text analysis, identifying, grouping, and reclassifying climate policies by peer countries, objectives, and similarities, enhancing policy comparison. [Python]

IMF Lending Programs Evaluation and Digitalization

Washington, DC

Part of IMF modernization project | [Brief](#)

Data Analyst

Jun 2021 – Feb 2023

- Compiled lending data (since 1952, resulting in 50,000+ rows) with key economic indicators and analytical groupings, analyzing countries' exposure and the impact of IMF lending policies. [SQL, Excel]
- Reconstructed and standardized the lending database using the IMF Monitoring of Fund Arrangements database, IMF Financial Data Query Tool, and internal data sources in SQL; cross-checked missing values by reviewing Staff Reports and resolved discrepancies. [SQL, Excel]
- Created an interactive prototype with automatic updates using multiple data sources for easier review, saving 95% of maintenance time. [Excel]
- Collaborated with the IT department to design and develop a dashboard for real-time monitoring and signaling of IMF lending program statuses, with weekly reports to senior IMF management. [SQL, PowerBI]

Using Sentiment to Measure Market Stress

Washington, DC

\$35,000 grant from International Monetary fund | [Brief](#)

Research Analyst & Programmer

Jun 2020 – May 2021

- Replicated financial risk prediction models based on IMF Working Paper WP/04/52, extended the ability of the existing EWS with a sentiment index constructed from news in Financial Times, and developed loops to automate the process. [Python, Excel, STATA]
- Backtested decision-tree, random-forest, and XGBoost model with monthly frequency data and did horse-racing comparisons using confusion matrix and F1 score, to match up financial crisis periods. [Python]

TEACHING EXPERIENCE

University of Georgia, Department of Agricultural and Applied Economics

- AAEC 2580 – Applied Microeconomic Principles | *Supervised by Dr. Greg Colson, Spring 2024*
- AAEC 4730 – World Food Economics | *Supervised by Dr. Ellen McCullough, Fall 2024*
- AAEC 3690 – Agribusiness Finance | *Supervised by Dr. Michael Adjemian, Spring 2025*